

Homework 13

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Practical 12: Survival Analysis

Answer to Problem 1

(a) *Is the Censoring Mechanism Uninformative?*

Let f_C and S_C denote the density function and survivor function of the censoring time C_i .

- If $\delta_i = 1$, then the event is observed at time y_i , meaning that $T_i = y_i, C_i \geq y_i$. Since T_i and C_i are independent, the joint contribution is $f_T(y_i; \lambda)S_C(y_i)$.
- If $\delta_i = 0$, then censoring occurs at time y_i , meaning that $C_i = y_i, T_i > y_i$. The joint contribution is $f_C(y_i)S_T(y_i; \lambda)$.

Thus, the likelihood is

$$L(\lambda, \eta) = \prod_{i=1}^n [f_T(y_i; \lambda)S_C(y_i; \eta)]^{\delta_i} [S_T(y_i; \lambda)f_C(y_i; \eta)]^{1-\delta_i},$$

where η denotes the parameters of the censoring distribution.

This can be factorized as

$$L(\lambda, \eta) = \left\{ \prod_{i=1}^n f_T(y_i; \lambda)^{\delta_i} S_T(y_i; \lambda)^{1-\delta_i} \right\} \left\{ \prod_{i=1}^n S_C(y_i; \eta)^{\delta_i} f_C(y_i; \eta)^{1-\delta_i} \right\}.$$

The first factor depends only on λ , while the second factor depends only on the censoring parameters η .

According to the condition given in the problem:

C_i is independent of T_i , and that the parameters of the distribution of C_i are distinct from the parameter λ

Therefore, the censoring mechanism is **uninformative** for inference about λ .

(b) MLE of λ and Its Asymptotic Distribution

Since $T_i \sim \text{Exp}(\lambda)$, we have

- $h(t) = \lambda$ and $H(t) = \lambda t$
- $S_T(t) = e^{-\lambda t}$ and $f_T(t) = \lambda e^{-\lambda t}$

The likelihood function is

$$\begin{aligned} L(\lambda) &= \prod_{i=1}^n f_T(y_i; \lambda)^{\delta_i} S_T(y_i; \lambda)^{1-\delta_i} \\ &= \prod_{i=1}^n [\lambda e^{-\lambda y_i}]^{\delta_i} [e^{-\lambda y_i}]^{1-\delta_i} \\ &= \prod_{i=1}^n \lambda^{\delta_i} e^{-\lambda y_i} \\ &= \lambda^{\sum_{i=1}^n \delta_i} \exp\left(-\lambda \sum_{i=1}^n y_i\right) \\ &= \lambda^D e^{-\lambda R} \end{aligned}$$

where $D = \sum_{i=1}^n \delta_i$ and $R = \sum_{i=1}^n y_i$.

The log-likelihood function is

$$\ell(\lambda) = D \log \lambda - \lambda R.$$

Differentiating with respect to λ gives

$$\frac{\partial \ell(\lambda)}{\partial \lambda} = \frac{D}{\lambda} - R.$$

Then we obtain

$$\boxed{\hat{\lambda}_{\text{MLE}} = \frac{D}{R} = \frac{\sum_{i=1}^n \delta_i}{\sum_{i=1}^n y_i}.$$

If $D = 0$, the likelihood is maximized in the limiting sense as $\hat{\lambda} \rightarrow 0$.

As for the asymptotic distribution, we use the corollary of MLE.

For one observation, the log-likelihood is

$$\ell_i(\lambda) = \delta_i \log \lambda - \lambda y_i.$$

The second derivative is

$$\frac{\partial^2 \ell_i(\lambda)}{\partial \lambda^2} = -\frac{\delta_i}{\lambda^2}.$$

Therefore, the Fisher information from one observation is

$$\begin{aligned} \mathcal{J}_1(\lambda) &= -E \left[\frac{\partial^2 \ell_i(\lambda)}{\partial \lambda^2} \right] \\ &= \frac{E(\Delta_i)}{\lambda^2}. \end{aligned}$$

Define

$$p = P(\Delta_i = 1) = P(T_i \leq C_i).$$

Then

$$\boxed{\mathcal{J}_1(\lambda) = \frac{p}{\lambda^2}.$$

Since T_i and C_i are independent,

$$\begin{aligned} p &= P(T_i \leq C_i) \\ &= \int_0^\infty f_T(t) P(C_i \geq t) dt \\ &= \int_0^\infty \lambda e^{-\lambda t} S_C(t) dt. \end{aligned}$$

By the asymptotic normality of the MLE,

$$\sqrt{n}(\hat{\lambda} - \lambda) \xrightarrow{d} N(0, \mathcal{J}_1(\lambda)^{-1}).$$

we obtain

$$\boxed{\sqrt{n}(\hat{\lambda} - \lambda) \xrightarrow{d} N\left(0, \frac{\lambda^2}{p}\right).$$

Equivalently,

$$\boxed{\hat{\lambda} \overset{a}{\sim} N\left(\lambda, \frac{\lambda^2}{np}\right).$$

Here $p = P(\Delta_i = 1)$ is the probability that the event is observed. Since the censoring distribution is not specified, p can not be reduced to a numerical value, and we estimate it by $\hat{p} = \frac{D}{n}$, where $D = \sum_{i=1}^n \delta_i$.

An estimated variance is

$$\widehat{\text{Var}}(\hat{\lambda}) \approx \frac{\hat{\lambda}^2}{D},$$

When there is no censoring, $p = 1$, and the result reduces to

$$\hat{\lambda} \stackrel{a}{\sim} N\left(\lambda, \frac{\lambda^2}{n}\right).$$

(c) the PDF and Survivor Function for T_i

It's easy to find that

$$H(t) = \begin{cases} \lambda_1 t, & 0 \leq t < \tau_1, \\ \lambda_1 \tau_1 + \lambda_2(t - \tau_1), & t \geq \tau_1. \end{cases}$$

Since

$$S_T(t) = \exp\{-H(t)\},$$

we have

$$S_T(t) = \begin{cases} \exp(-\lambda_1 t), & 0 \leq t < \tau_1, \\ \exp[-\lambda_1 \tau_1 - \lambda_2(t - \tau_1)], & t \geq \tau_1. \end{cases}$$

Since $f_T(t) = h(t)S_T(t)$, we obtain

$$f_T(t) = \begin{cases} \lambda_1 \exp(-\lambda_1 t), & 0 \leq t < \tau_1, \\ \lambda_2 \exp[-\lambda_1 \tau_1 - \lambda_2(t - \tau_1)], & t \geq \tau_1. \end{cases}$$

Equivalently, in terms of (ν_1, ν_2) ,

$$f_T(t) = \begin{cases} e^{\nu_1} \exp(-e^{\nu_1} t), & 0 \leq t < \tau_1, \\ e^{\nu_2} \exp[-e^{\nu_1} \tau_1 - e^{\nu_2}(t - \tau_1)], & t \geq \tau_1. \end{cases}$$

(d) *MLE of (ν_1, ν_2) and Its Asymptotic Distribution*

Define

$$I_{ij} = \begin{cases} 1, & \tau_{j-1} \leq y_i < \tau_j, \\ 0, & \text{otherwise,} \end{cases} \quad j = 1, 2.$$

which indicates whether the observed time y_i falls in interval j .

For the two intervals,

$$I_{i1} = I(0 \leq y_i < \tau_1), \quad I_{i2} = I(y_i \geq \tau_1).$$

Define the time spent by individual i in interval j as

$$y_{ij} = \max \{0, \min(\tau_j - \tau_{j-1}, y_i - \tau_{j-1})\}.$$

In the present two-interval case,

$$y_{i1} = \min(y_i, \tau_1)$$

and

$$y_{i2} = (y_i - \tau_1)_+ = \max(0, y_i - \tau_1).$$

Now define

$$d_j = \sum_{i=1}^n \delta_i I_{ij}, \quad r_j = \sum_{i=1}^n y_{ij}, \quad j = 1, 2.$$

Here,

- d_j is the number of observed events in interval j
- r_j is the total time at risk contributed in interval j

For individual i , the likelihood related to the survival parameters is

$$L_i(\nu_1, \nu_2) = h(y_i)^{\delta_i} \exp\{-H(y_i)\}.$$

Using the interval notation, this can be written as

$$L_i(\nu_1, \nu_2) = \prod_{j=1}^2 (e^{\nu_j})^{\delta_i I_{ij}} \exp(-e^{\nu_j} y_{ij}).$$

Therefore, the likelihood function is

$$\begin{aligned} L(\nu_1, \nu_2) &= \prod_{i=1}^n \prod_{j=1}^2 (e^{\nu_j})^{\delta_i I_{ij}} \exp(-e^{\nu_j} y_{ij}) \\ &= \prod_{j=1}^2 (e^{\nu_j})^{d_j} \exp(-e^{\nu_j} r_j). \end{aligned}$$

Hence, the log-likelihood function is

$$\ell(\nu_1, \nu_2) = d_1 \nu_1 - r_1 e^{\nu_1} + d_2 \nu_2 - r_2 e^{\nu_2}.$$

For $j = 1, 2$,

$$\frac{\partial \ell}{\partial \nu_j} = d_j - r_j e^{\nu_j}.$$

Therefore,

$$e^{\hat{\nu}_j} = \frac{d_j}{r_j}.$$

Since $\lambda_j = e^{\nu_j}$, we obtain the MLEs

$$\hat{\lambda}_j = \frac{d_j}{r_j}, \hat{\nu}_j = \log\left(\frac{d_j}{r_j}\right), j = 1, 2.$$

That is,

$$\hat{\nu}_1 = \log\left(\frac{\sum_{i=1}^n \delta_i I_{i1}}{\sum_{i=1}^n y_{i1}}\right), \hat{\nu}_2 = \log\left(\frac{\sum_{i=1}^n \delta_i I_{i2}}{\sum_{i=1}^n y_{i2}}\right).$$

If $d_j = 0$ for some interval j , then $\hat{\lambda}_j = 0$ and $\hat{\nu}_j = -\infty$. Thus, there is no finite interior MLE for ν_j when no observed event occurs in interval j .

The second derivatives of the log-likelihood are

$$\frac{\partial^2 \ell}{\partial \nu_j^2} = -r_j e^{\nu_j}, \quad j = 1, 2,$$

and

$$\frac{\partial^2 \ell}{\partial \nu_1 \partial \nu_2} = 0.$$

For one observation, define

$$m_j = E(Y_{ij}), \quad j = 1, 2.$$

Then the Fisher information contribution for ν_j is

$$\mathcal{J}_j = e^{\nu_j} m_j.$$

Define

$$\pi_j = P(\Delta_i = 1, \tau_{j-1} \leq Y_i < \tau_j), \quad j = 1, 2.$$

This is the probability that an individual has an observed event in interval j .

Under independent censoring, $\pi_j = e^{\nu_j} E(Y_{ij})$, the Fisher information matrix for one observation is

$$\mathcal{J}(\nu_1, \nu_2) = \begin{pmatrix} \pi_1 & 0 \\ 0 & \pi_2 \end{pmatrix}$$

Where

$$\pi_1 = \int_0^{\tau_1} e^{\nu_1} \exp(-e^{\nu_1} t) S_C(t) dt, \pi_2 = \int_{\tau_1}^{\infty} e^{\nu_2} \exp[-e^{\nu_1} \tau_1 - e^{\nu_2} (t - \tau_1)] S_C(t) dt.$$

Assume that $\pi_1 > 0$ and $\pi_2 > 0$.

Then, by the asymptotic normality of the MLE,

$$\boxed{\sqrt{n} \begin{pmatrix} \hat{\nu}_1 - \nu_1 \\ \hat{\nu}_2 - \nu_2 \end{pmatrix} \xrightarrow{d} N_2 \left[\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1/\pi_1 & 0 \\ 0 & 1/\pi_2 \end{pmatrix} \right].}$$

Equivalently, for each $j = 1, 2$,

$$\boxed{\hat{\nu}_j \stackrel{a}{\sim} N \left(\nu_j, \frac{1}{n\pi_j} \right).}$$

Since $\hat{\pi}_j = \frac{d_j}{n}$, we may estimate the asymptotic variance by $\widehat{\text{Var}}(\hat{\nu}_j) \approx \frac{1}{d_j}$.

As for *why use the parameterization* $\lambda_j = e^{\nu_j}$, the hazard rates must satisfy $\lambda_j > 0$. If we estimate λ_j directly, the optimization problem must be constrained to positive values.

By writing $\lambda_j = e^{\nu_j}$, we allow $\nu_j \in \mathbb{R}$ without any additional constraints, while automatically ensuring that $\lambda_j > 0$.

This parameterization also gives a convenient log-likelihood:

$$\ell(\nu_1, \nu_2) = \sum_{j=1}^2 [d_j \nu_j - r_j e^{\nu_j}] ,$$

which is concave in (ν_1, ν_2) .

Additional question: Is T_i being independent of C_i sufficient for censoring to be uninformative?

(Consider the situation where T_i is independent of C_i , but both T_i and C_i are distributed as Exponential random variables with common rate parameter λ .)

NOT SUFFICIENT.

Under independent censoring, the full likelihood can be factorized as

$$L(\lambda, \eta) = \left\{ \prod_{i=1}^n f_T(y_i; \lambda)^{\delta_i} S_T(y_i; \lambda)^{1-\delta_i} \right\} \left\{ \prod_{i=1}^n S_C(y_i; \eta)^{\delta_i} f_C(y_i; \eta)^{1-\delta_i} \right\} .$$

If the censoring parameter η is distinct from λ , then the second factor does not involve λ . Therefore, when estimating λ , it can be ignored. In this case, the censoring mechanism is uninformative.

However, suppose that $T_i \sim \text{Exp}(\lambda)$, $C_i \sim \text{Exp}(\lambda)$ even though T_i and C_i are independent. Then the censoring also depends on λ .

The full likelihood becomes

$$L(\lambda) = \prod_{i=1}^n [f_T(y_i; \lambda) S_C(y_i; \lambda)]^{\delta_i} [S_T(y_i; \lambda) f_C(y_i; \lambda)]^{1-\delta_i} .$$

Since $f_T(y; \lambda) = \lambda e^{-\lambda y}$, $S_T(y; \lambda) = e^{-\lambda y}$,

we get

$$L(\lambda) = \lambda^n \exp \left(-2\lambda \sum_{i=1}^n y_i \right) .$$

Thus the full-likelihood MLE is

$$\hat{\lambda} = \frac{n}{2 \sum_{i=1}^n y_i}.$$

But if we incorrectly ignore the censoring, we would use only

$$L_T(\lambda) = \prod_{i=1}^n f_T(y_i; \lambda)^{\delta_i} S_T(y_i; \lambda)^{1-\delta_i} = \lambda^{\sum_i \delta_i} \exp \left(-\lambda \sum_i y_i \right),$$

which gives

$$\hat{\lambda} = \frac{\sum_{i=1}^n \delta_i}{\sum_{i=1}^n y_i}.$$

These two estimators are generally different.

Problem 2

```
# Load the file Leuk.csv into R
Leuk <- read.csv("Leuk.csv")
# 0 = Placebo, 1 = 6-MP
Leuk$drug_group <- factor(
  Leuk$drug,
  levels = c(0, 1),
  labels = c("Placebo", "6-MP")
)
# 0 = Female, 1 = Male
Leuk$sex_group <- factor(
  Leuk$sex,
  levels = c(0, 1),
  labels = c("Female", "Male")
)
```

```
# Overview of the data
head(Leuk)
  time drug cens logwbc sex drug_group sex_group
1    1    0    1   2.80  1   Placebo      Male
2    1    0    1   5.00  1   Placebo      Male
3    2    0    1   4.91  1   Placebo      Male
```

4	2	0	1	4.48	1	Placebo	Male
5	3	0	1	4.01	1	Placebo	Male
6	4	0	1	4.36	1	Placebo	Male

```
summary(Leuk)
```

time		drug		cens		logwbc	
Min.	: 1.00	Min.	:0.0	Min.	:0.0000	Min.	:1.450
1st Qu.	: 6.00	1st Qu.	:0.0	1st Qu.	:0.0000	1st Qu.	:2.303
Median	:10.50	Median	:0.5	Median	:1.0000	Median	:2.800
Mean	:12.88	Mean	:0.5	Mean	:0.7143	Mean	:2.930
3rd Qu.	:18.50	3rd Qu.	:1.0	3rd Qu.	:1.0000	3rd Qu.	:3.490
Max.	:35.00	Max.	:1.0	Max.	:1.0000	Max.	:5.000

sex		drug_group		sex_group	
Min.	:0.0000	Placebo	:21	Female	:22
1st Qu.	:0.0000	6-MP	:21	Male	:20
Median	:0.0000				
Mean	:0.4762				
3rd Qu.	:1.0000				
Max.	:1.0000				

(a) *Do you observe any initial ‘differences’ between the two treatment groups?*

```
# Comparison of 2 groups
# time and logwbc are continuous
aggregate(
  cbind(time,logwbc) ~ drug_group,
  data = Leuk,
  FUN = function(x) c(mean = mean(x),sd = sd(x))
)
drug_group time.mean time.sd logwbc.mean logwbc.sd
1 Placebo 8.666667 6.467869 3.2242857 0.9722786
2 6-MP 17.095238 9.999524 2.6361905 0.7738765
```

```
# cens and sex_group are categorical
# 0 = censoring, 1 = coming out of remission
with(Leuk, list(event = table(drug_group, cens),sex = table(drug_group, sex_group)))
$event
```

```

      cens
drug_group 0  1
  Placebo  0 21
    6-MP   12  9

$sex
      sex_group
drug_group Female Male
  Placebo      11    10
    6-MP       11    10

```

- The 6-MP group has a **longer** average remission *time*, about 17.10 weeks, while the placebo group has an average remission time of about 8.67 weeks.
- **All** placebo patients came out of remission, but many 6-MP patients were **censored**, which suggests longer remission under 6-MP.
- The *sex* distribution is **balanced** between the two groups.

```

leuk.survobj <- Surv(time = Leuk$time, event = Leuk$cens) # 创建生存对象
# 没有加号表示事件发生, 加号表示右删失
leuk.survobj
[1]  1   1   2   2   3   4   4   5   5   8   8   8   8  11  11  12  12  15  17
[20] 22  23  6+  6   6   6   7   9+ 10+ 10  11+ 13  16  17+ 19+ 20+ 22  23  25+
[39] 32+ 32+ 34+ 35+

```

Kaplan-Meier Curves

```

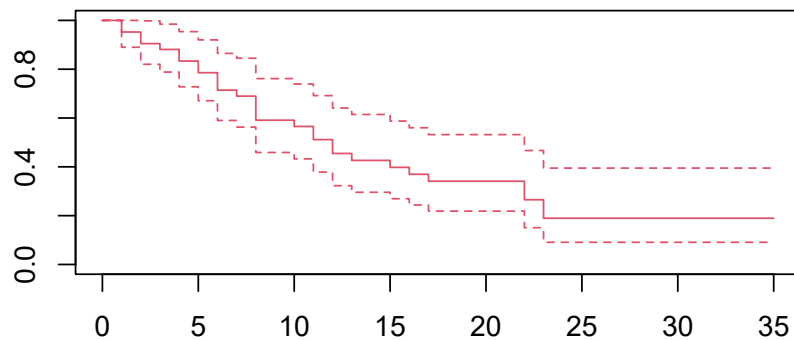
# obtain the overall K-M curve
leuk.kmoverall <- survfit(leuk.survobj ~ 1, data = Leuk)
# n.risk: 仍在观察中 survival: 生存概率
print_no_call(summary(leuk.kmoverall))

```

time	n.risk	n.event	survival	std.err	lower 95% CI	upper 95% CI
1	42	2	0.952	0.0329	0.8901	1.000
2	40	2	0.905	0.0453	0.8202	0.998
3	38	1	0.881	0.0500	0.7883	0.985
4	37	2	0.833	0.0575	0.7279	0.954
5	35	2	0.786	0.0633	0.6709	0.920

6	33	3	0.714	0.0697	0.5899	0.865
7	29	1	0.690	0.0715	0.5628	0.845
8	28	4	0.591	0.0764	0.4588	0.762
10	23	1	0.565	0.0773	0.4325	0.739
11	21	2	0.512	0.0788	0.3783	0.692
12	18	2	0.455	0.0796	0.3227	0.641
13	16	1	0.426	0.0795	0.2958	0.615
15	15	1	0.398	0.0791	0.2694	0.588
16	14	1	0.369	0.0784	0.2437	0.560
17	13	1	0.341	0.0774	0.2186	0.532
22	9	2	0.265	0.0765	0.1507	0.467
23	7	2	0.189	0.0710	0.0909	0.395

Overall K-M curve



```
# obtain K-M curves for the treatment groups
leuk.kmdrug <- survfit(leuk.survobj ~ drug_group, data = Leuk)
knitr::kable(leuk.kmdrug$table)
```

```
print_no_call(summary(leuk.kmdrug)) # summary(leuk.kmdrug)
```

drug_group=Placebo						
time	n.risk	n.event	survival	std.err	lower 95% CI	upper 95% CI
1	21	2	0.9048	0.0641	0.78754	1.000
2	19	2	0.8095	0.0857	0.65785	0.996
3	17	1	0.7619	0.0929	0.59988	0.968
4	16	2	0.6667	0.1029	0.49268	0.902

5	14	2	0.5714	0.1080	0.39455	0.828
8	12	4	0.3810	0.1060	0.22085	0.657
11	8	2	0.2857	0.0986	0.14529	0.562
12	6	2	0.1905	0.0857	0.07887	0.460
15	4	1	0.1429	0.0764	0.05011	0.407
17	3	1	0.0952	0.0641	0.02549	0.356
22	2	1	0.0476	0.0465	0.00703	0.322
23	1	1	0.0000	NaN	NA	NA

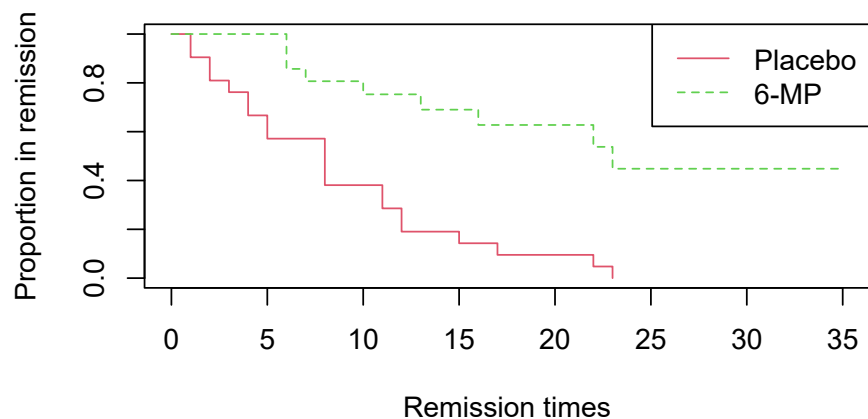
drug_group=6-MP

time	n.risk	n.event	survival	std.err	lower 95% CI	upper 95% CI
6	21	3	0.857	0.0764	0.720	1.000
7	17	1	0.807	0.0869	0.653	0.996
10	15	1	0.753	0.0963	0.586	0.968
13	12	1	0.690	0.1068	0.510	0.935
16	11	1	0.627	0.1141	0.439	0.896
22	7	1	0.538	0.1282	0.337	0.858
23	6	1	0.448	0.1346	0.249	0.807

```
knitr::kable(km23_table, digits = 3) # summary(leuk.kmdrug, times = 23)
```

group	time	risk	event	surv	se	lower_CI	upper_CI
Placebo	23	1	21	0.000	NaN	NA	NA
6-MP	23	6	9	0.448	0.135	0.249	0.807

K-M curves for the two treatment groups

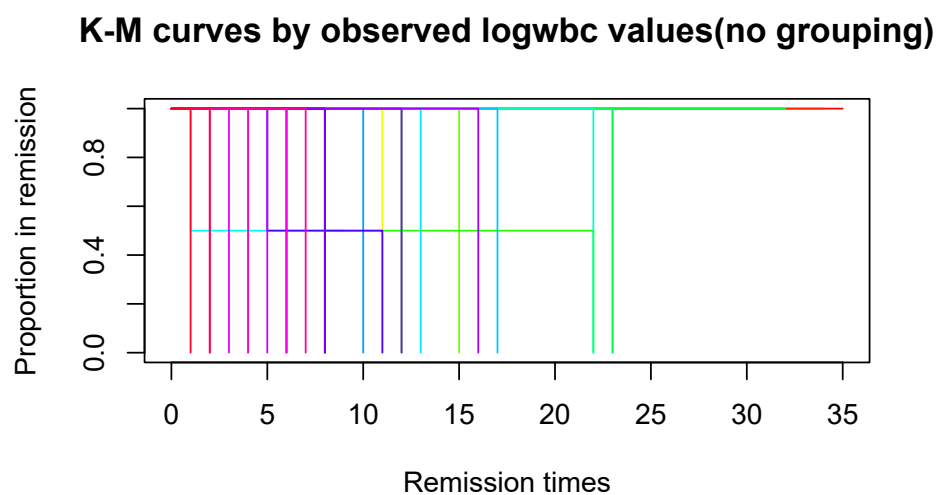
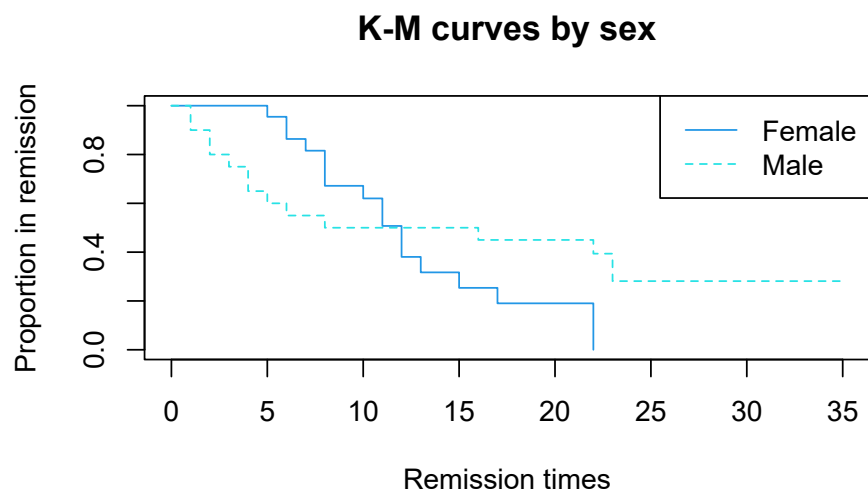


(b) *What are these curves telling us?*

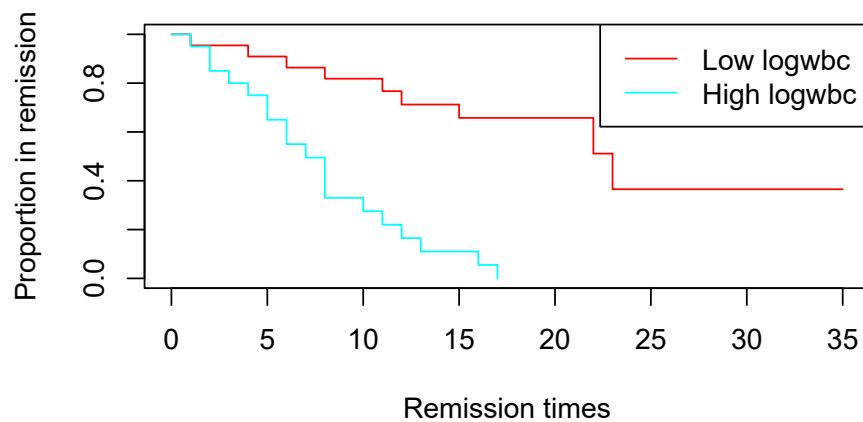
- The K-M curves show that patients receiving 6-MP generally remain in remission **longer** than patients receiving placebo.
- At week 23, the estimated survivor probability is about 0 for placebo and about 0.45 for 6-MP.

(c) *Now plot the survival curves by sex and by logwbc.*

(logwbc is continuous, should we group it ?)



K-M curves by observed logwbc values(grouped)



Log-Rank Tests

(to assess whether patients in the 6-MP group remaining generally in remission longer)

```
logrank.drug <- survdiff(leuk.survobj ~ drug_group, data = Leuk)
print_no_call(logrank.drug)
```

	N	Observed	Expected	$(O-E)^2/E$	$(O-E)^2/V$
drug_group=Placebo	21	21	10.7	9.77	16.8
drug_group=6-MP	21	9	19.3	5.46	16.8

Chisq= 16.8 on 1 degrees of freedom, p= 4e-05

(d) *What do you conclude? What are the test-statistic values for the log-rank statistic and its approximation?*

- The drug log-rank test gives a **chi-square statistic** of about 16.8 with 1 degree of freedom and **p-value** about $4e - 05$.
- There is strong evidence that the remission distributions **differ** between placebo and 6-MP.

(e) *Perform now the log-rank test on sex and logwbc. What are your conclusions? Was it appropriate to use the log-rank test for these variables?*

```
# log-rank tests on sex
logrank.sex <- survdiff(leuk.survobj ~ sex_group, data = Leuk)
print_no_call(logrank.sex)
```

	N	Observed	Expected	(O-E) ² /E	(O-E) ² /V
sex_group=Female	22	16	14.2	0.240	0.557
sex_group=Male	20	14	15.8	0.214	0.557

Chisq= 0.6 on 1 degrees of freedom, p= 0.5

```
# log-rank tests on continuous logwbc
#logrank.logwbc <- survdiff(leuk.survobj ~ logwbc, data = Leuk)

# log-rank tests on grouped logwbc
logrank.logwbc <- survdiff(leuk.survobj ~ logwbc_group, data = Leuk)
print_no_call(logrank.logwbc)
```

	N	Observed	Expected	(O-E) ² /E	(O-E) ² /V
logwbc_group=Low logwbc	22	11	21.39	5.05	20.9
logwbc_group=High logwbc	20	19	8.61	12.55	20.9

Chisq= 20.9 on 1 degrees of freedom, p= 5e-06

- The sex log-rank test is *not* significant, with **chi-square** about 0.6 and **p-value** about 0.5.
- The log-rank test is *not* really appropriate for raw continuous logwbc so that we **grouped** logwbc into 2 groups for log-rank tests.
- After grouping the continuous variable logwbc into low and high groups, the log-rank test shows a highly **significant** difference between the two survival curves.
- The low-logwbc group has 11 observed events, much fewer than the 21.39 events expected under the null hypothesis of equal survival curves.
- The high-logwbc group has 19 observed events, much more than the 8.61 expected events.
- The log-rank **chi-square** statistic is 20.9 with 1 degree of freedom, and the **p-value** is about 5×10^{-6} . Therefore, we **reject** the null hypothesis and conclude that remission times **differ significantly** between the two logwbc groups.

- Patients with higher logwbc tend to come out of remission earlier, while patients with lower logwbc tend to remain in remission longer.

Cox Proportional Hazards Models

To assess the effect of 6-MP on the hazard of coming out of remission, we begin by only including drug into the Cox PH model (i.e. fitting an unadjusted Cox PH model).

We fit the Cox proportional hazards model

$$h(t \mid \text{drug}) = h_0(t) \exp(\beta \cdot \text{drug}),$$

where:

- $h(t \mid \text{drug})$ is the hazard of relapse at time t
- $h_0(t)$ is the baseline hazard for the placebo group
- $\text{drug} = 0$ represents the placebo group, $\text{drug} = 1$ represents the 6-MP treatment group
- β measures the treatment effect

```
leuk.ph <- coxph(leuk.survobj ~ drug, data = Leuk)
print_no_call(summary(leuk.ph))

n= 42, number of events= 30

      coef exp(coef) se(coef)      z Pr(>|z|)
drug -1.5721    0.2076    0.4124 -3.812 0.000138 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

      exp(coef) exp(-coef) lower .95 upper .95
drug    0.2076      4.817    0.09251    0.4659

Concordance= 0.69 (se = 0.041 )
Likelihood ratio test= 16.35 on 1 df,  p=5e-05
Wald test              = 14.53 on 1 df,  p=1e-04
Score (logrank) test = 17.25 on 1 df,  p=3e-05
```

(f) *Do you understand the output? How do you interpret the drug effect? What hypothesis is the likelihood ratio and Wald tests in the output testing?*

The fitted model is

$$\hat{h}(t \mid \text{drug}) = h_0(t) \exp(-1.5721 \cdot \text{drug}).$$

Thus, the estimated hazard ratio for 6-MP versus placebo is

$$\exp(\hat{\beta}) = \exp(-1.5721) = 0.2076.$$

- The hazard ratio is about **0.208**, meaning that the relapse hazard for patients receiving 6-MP is approximately 20.8% of that for patients receiving placebo.
- The p-value for the drug coefficient is $p = 0.000138$, which is much smaller than 0.05. Therefore, the treatment effect is statistically **significant**.
- The 95% confidence interval for the hazard ratio is (0.0925, 0.4659), which is entirely **below** 1. This also indicates that 6-MP significantly **lowers** the risk of relapse compared with placebo.
- Overall, the Cox model suggests that patients treated with 6-MP tend to remain in remission **longer** than patients in the placebo group.
- The likelihood ratio and Wald tests are testing the null hypothesis that the *drug coefficient is zero*.

(g) *Based on the variables provided in Leuk, come up with a “best” fitting Cox PH model. Is there a drug by logwbc interaction? Interpret the results from your best/final model.*

```
leuk.ph.drug.logwbc <- coxph(leuk.survobj ~ drug + logwbc, data = Leuk)
leuk.ph.full <- coxph(leuk.survobj ~ drug + sex + logwbc, data = Leuk)
leuk.ph.int <- coxph(leuk.survobj ~ drug * logwbc + sex, data = Leuk)
anova(leuk.ph.full, leuk.ph.int, test = "Chisq") # 检验交互项
```

Analysis of Deviance Table

	Cox model:	response is	leuk.survobj
Model 1:	~ drug + sex + logwbc		
Model 2:	~ drug * logwbc + sex		
	loglik	Chisq	Df Pr(> Chi)
1	-69.590		
2	-69.326	0.5286	1 0.4672

```
print_no_call(summary(leuk.ph.drug.logwbc)) # drug + logwbc
```

n= 42, number of events= 30

	coef	exp(coef)	se(coef)	z	Pr(> z)	
drug	-1.3861	0.2501	0.4248	-3.263	0.0011	**
logwbc	1.6909	5.4243	0.3359	5.034	4.8e-07	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

	exp(coef)	exp(-coef)	lower .95	upper .95
drug	0.2501	3.9991	0.1088	0.5749
logwbc	5.4243	0.1844	2.8082	10.4776

Concordance= 0.852 (se = 0.04)
Likelihood ratio test= 46.71 on 2 df, p=7e-11
Wald test = 33.6 on 2 df, p=5e-08
Score (logrank) test = 46.07 on 2 df, p=1e-10

```
print_no_call(summary(leuk.ph.full)) # drug + sex + logwbc
```

n= 42, number of events= 30

	coef	exp(coef)	se(coef)	z	Pr(> z)	
drug	-1.5036	0.2223	0.4615	-3.258	0.00112	**
sex	0.3147	1.3698	0.4545	0.692	0.48872	
logwbc	1.6819	5.3760	0.3366	4.997	5.82e-07	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

	exp(coef)	exp(-coef)	lower .95	upper .95
drug	0.2223	4.498	0.08998	0.5493
sex	1.3698	0.730	0.56206	3.3384
logwbc	5.3760	0.186	2.77944	10.3982

Concordance= 0.851 (se = 0.041)
Likelihood ratio test= 47.19 on 3 df, p=3e-10
Wald test = 33.54 on 3 df, p=2e-07
Score (logrank) test = 48.01 on 3 df, p=2e-10

```
print_no_call(summary(leuk.ph.int)) # drug * logwbc + sex
```

n= 42, number of events= 30

	coef	exp(coef)	se(coef)	z	Pr(> z)
drug	-2.7303	0.0652	1.7516	-1.559	0.119056
logwbc	1.5183	4.5644	0.3961	3.833	0.000127 ***
sex	0.3635	1.4383	0.4487	0.810	0.417942
drug:logwbc	0.3910	1.4784	0.5336	0.733	0.463709

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

	exp(coef)	exp(-coef)	lower .95	upper .95
drug	0.0652	15.3379	0.002105	2.019
logwbc	4.5644	0.2191	2.100055	9.921
sex	1.4383	0.6953	0.596884	3.466
drug:logwbc	1.4784	0.6764	0.519535	4.207

Concordance= 0.858 (se = 0.04)
Likelihood ratio test= 47.72 on 4 df, p=1e-09
Wald test = 32.14 on 4 df, p=2e-06
Score (logrank) test = 51.04 on 4 df, p=2e-10

```
AIC(leuk.ph,leuk.ph.drug.logwbc,leuk.ph.full,leuk.ph.int) # 比较各个模型的 AIC
```

	df	AIC
leuk.ph	1	172.0168
leuk.ph.drug.logwbc	2	143.6562
leuk.ph.full	3	145.1810
leuk.ph.int	4	146.6523

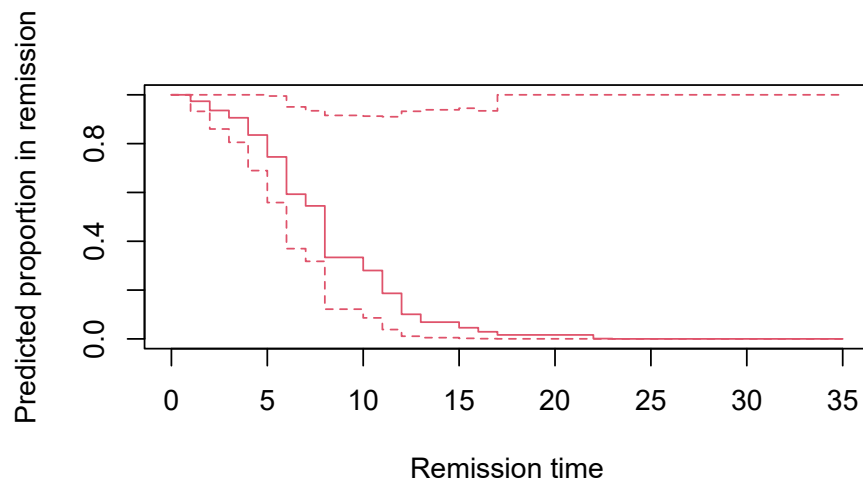
- The drug-by-logwbc interaction is *not* statistically significant, with a p-value of about 0.464.
- We also compared the additive Cox model with the interaction model using a likelihood ratio test. The test statistic is 0.5286 with 1 degree of freedom, and the **p-value** is 0.4672, which is much larger than 0.05.
- Among the fitted Cox models, the model **including drug and continuous logwbc** has the smallest AIC, so it is selected as the final model.
- In this final model, the **hazard ratio** for 6-MP versus placebo is 0.250, meaning that 6-MP reduces the relapse hazard by about 75% after adjusting for logwbc.

- For logwbc, the hazard ratio is 5.424 per one-unit increase, suggesting that higher logwbc is strongly associated with earlier relapse.

(h) *Can you figure out how to plot the predicted survival curve for a person who is on 6-MP and has a logwbc value of 4.00? You may want to look at the help file for the R command `survfit`.*

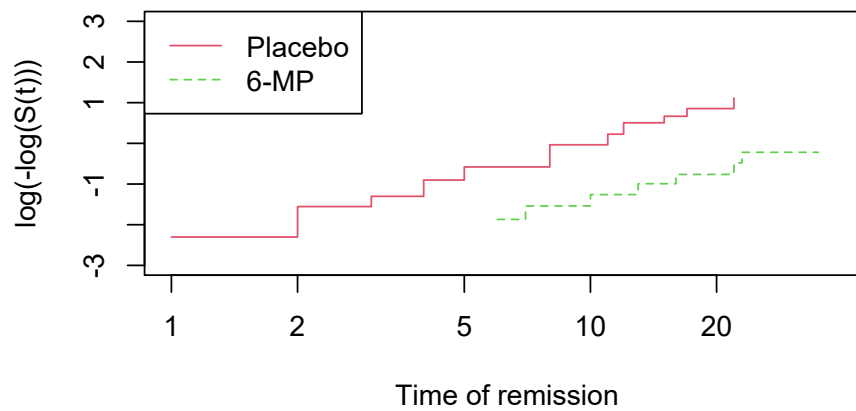
```
new.patient <- data.frame(drug = 1, logwbc = 4.00)
new.patient.surv <- survfit(leuk.ph.drug.logwbc, newdata = new.patient)
print_no_call(new.patient.surv)
```

```
      n events median 0.95LCL 0.95UCL
[1,] 42      30      8       6      NA
```



Assessing the Proportional Hazards Assumption

To assess the validity of the PH assumption for the drug variable, begin by plotting the complementary log-log curves for the Placebo and 6-MP groups.

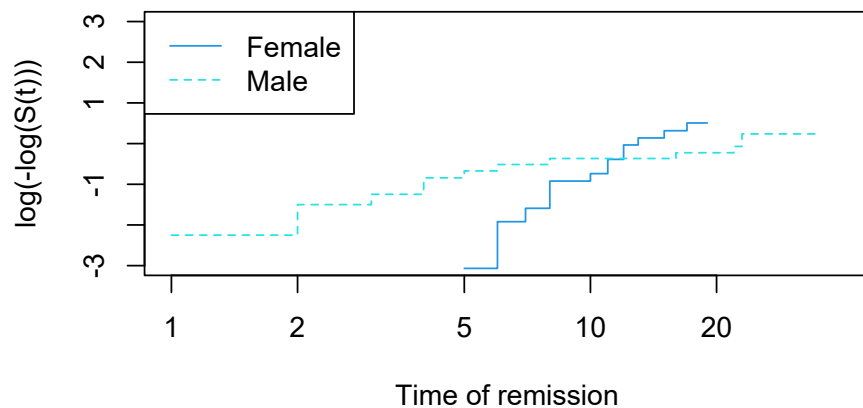


```
cox.zph(leuk.ph) # using the default K-M transformation of time
      chisq df    p
drug    0.0229  1 0.88
GLOBAL  0.0229  1 0.88
```

(i) *What do you conclude?*

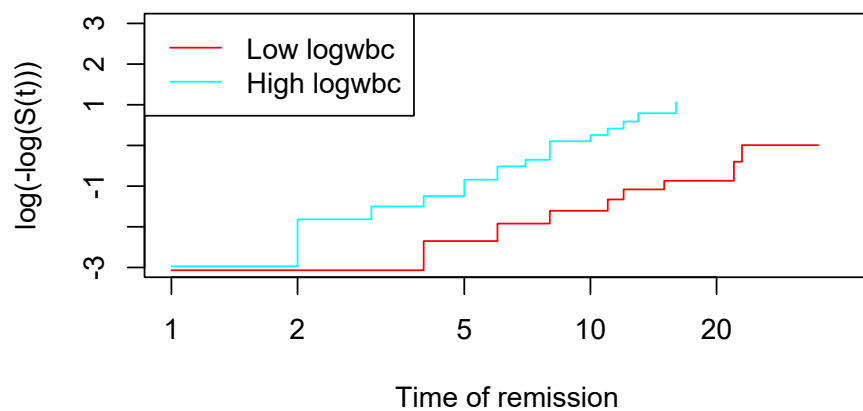
- The complementary log-log curves do *not* show a major violation for drug.
- The Schoenfeld residual test gives a p-value about 0.88 for drug, so there is *no* evidence against the proportional hazards assumption for the drug variable in the unadjusted Cox model.

(j) Test whether the PH assumption is satisfied for sex and logwbc.



```
cox.zph(leuk.ph.full)
      chisq df    p
drug   0.036  1 0.85
sex    5.420  1 0.02
logwbc 0.142  1 0.71
GLOBAL 5.879  3 0.12
```

(For *logwbc*, which is a continuous variable, the complementary log-log plot should be based on grouped values rather than treating every observed value as a separate group.)



```
# check the selected Cox model with drug and logwbc
```

```
cox.zph(leuk.ph.drug.logwbc)
```

	chisq	df	p
drug	8.27e-05	1	0.99
logwbc	4.00e-02	1	0.84
GLOBAL	4.02e-02	2	0.98

- In the full Cox model including **drug**, **sex**, and **logwbc**, the Schoenfeld residual test gives a **p-value** for **sex** about 0.02, suggesting a possible **violation** of the proportional hazards assumption for **sex**.
- In the final selected Cox model with **drug** and **logwbc**, the Schoenfeld residual test does *not* show evidence of PH violation for either variable. The **p-values** are about 0.99 for **drug**, 0.84 for **logwbc**, and 0.98 for the global test.
- As for the full model where **sex** is included, a reasonable approach is to stratify by **sex**.
- In the stratified Cox model, the PH tests for **drug** and **logwbc** are *not* significant, so the stratified model also appears appropriate.

(If the PH assumption is not satisfied, re-fit the Cox model but now stratifying by the variable not satisfying the PH assumption.)

```
leuk.sc <- coxph(leuk.survobj ~ drug + logwbc + strata(sex), data = Leuk)
```

```
print_no_call(summary(leuk.sc))
```

```
n= 42, number of events= 30
```

	coef	exp(coef)	se(coef)	z	Pr(> z)
drug	-0.9981	0.3686	0.4736	-2.108	0.0351 *
logwbc	1.4537	4.2787	0.3441	4.225	2.39e-05 ***

```
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

	exp(coef)	exp(-coef)	lower .95	upper .95
drug	0.3686	2.7131	0.1457	0.9324
logwbc	4.2787	0.2337	2.1799	8.3982

```
Concordance= 0.812 (se = 0.059 )
```

```
Likelihood ratio test= 32.06 on 2 df, p=1e-07
```

```
Wald test = 22.75 on 2 df, p=1e-05
```

```
Score (logrank) test = 30.8 on 2 df, p=2e-07
```

```
cox.zph(leuk.sc)
```

	chisq	df	p
drug	0.148	1	0.70
logwbc	0.318	1	0.57
GLOBAL	0.478	2	0.79

Problem 3

```
# load the data
lungdat <- survival::lung
# 0 = censored, 1 = dead
lungdat$status <- lungdat$status - 1 # creating 0/1 event indicator
# ECOG 0/ECOG 1/ECOG 2+
lungdat$ph.ecogcat <- ifelse(lungdat$ph.ecog >= 2, 2, lungdat$ph.ecog)
# wt.loss = weight loss in last six months; positive values mean weight loss
lungdat$wtlossyn <- ifelse(lungdat$wt.loss > 0, 1, 0)
head(lungdat)
```

	inst	time	status	age	sex	ph.ecog	ph.karno	pat.karno	meal.cal	wt.loss
1	3	306	1	74	1	1	90	100	1175	NA
2	3	455	1	68	1	0	90	90	1225	15
3	3	1010	0	56	1	0	90	90	NA	15
4	5	210	1	57	1	1	90	60	1150	11
5	1	883	1	60	1	0	100	90	NA	0
6	12	1022	0	74	1	1	50	80	513	0

```
ph.ecogcat wtlossyn
1          1      NA
2          0       1
3          0       1
4          1       1
5          0       0
6          1       0
```

(a) *What does the overall survival distribution look like?*

```
lung.survobj <- Surv(time = lungdat$time, event = lungdat$status) # 创建生存对象
lung.kmoverall <- survfit(lung.survobj ~ 1, data = lungdat) # 包括所有病人，不分组
```

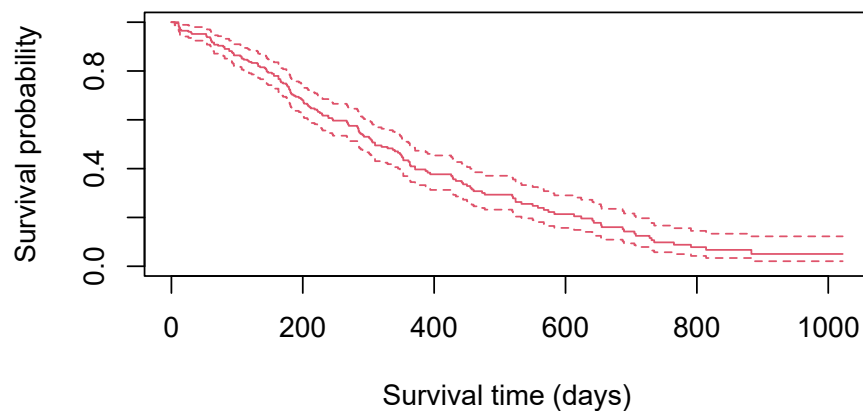
```
print_no_call(lung.kmoverall)
```

```
      n events median 0.95LCL 0.95UCL
[1,] 228    165    310    285    363
```

```
print_no_call(summary(lung.kmoverall, times = c(180, 365, 730))) # 挑 3 个时间点
```

time	n.risk	n.event	survival	std.err	lower 95% CI	upper 95% CI
180	160	63	0.722	0.0298	0.6655	0.783
365	65	58	0.409	0.0358	0.3447	0.486
730	13	38	0.116	0.0283	0.0716	0.187

Overall K-M curve for the lung cancer data



- The overall survival curve **decreases** steadily over time.
- The **median** survival time is about 310 days
- The estimated **survival probability** at one year is roughly 0.41

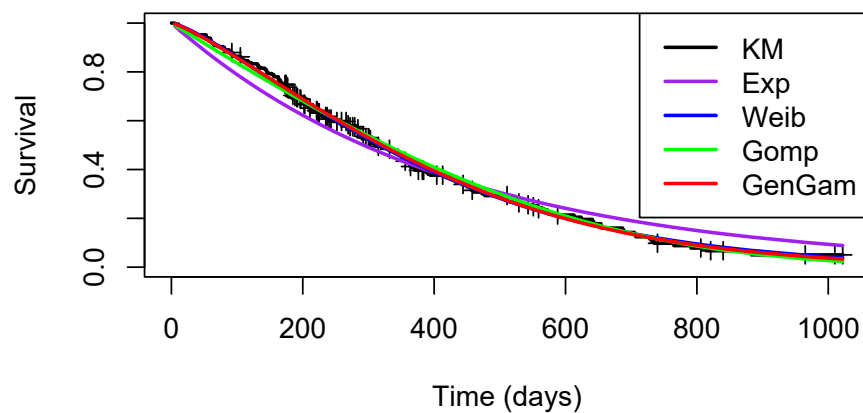
(b) *Which of these four distributions is preferred? Why?*

```
ex <- flexsurvreg(Surv(time, status) ~ 1, data = lungdat, dist = "exponential")
we <- flexsurvreg(Surv(time, status) ~ 1, data = lungdat, dist = "weibull")
go <- flexsurvreg(Surv(time, status) ~ 1, data = lungdat, dist = "gompertz")
gg <- flexsurvreg(Surv(time, status) ~ 1, data = lungdat, dist = "gengamma")
```

	distribution	logLik	AIC
2	Weibull	-1153.851	2311.702
4	GenGamma	-1153.690	2313.380
3	Gompertz	-1155.355	2314.711
1	Exponential	-1162.338	2326.676

- The **GenGamma** distribution has the highest log-likelihood, but the log-likelihood of the **Weibull** distribution is very close to it.
- A **smaller** AIC indicates a better balance between model fit and model complexity. As a result, the **Weibull** distribution provides the best fit among the four parametric models.

(c) *What do you conclude from this figure?*

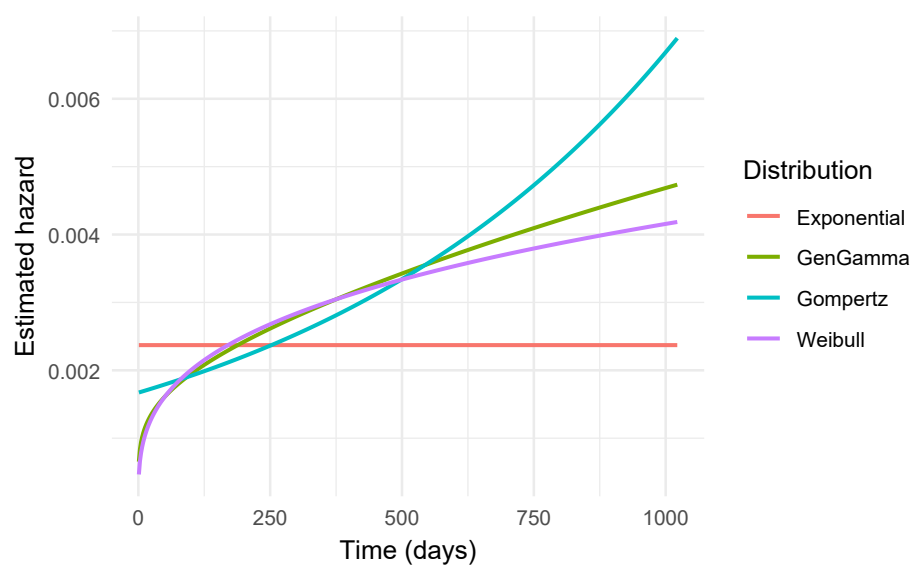


- The **exponential** model appears to provide a poorer fit to the Kaplan-Meier curve.
- It tends to underestimate survival at earlier times and overestimate survival at later times, suggesting that the constant hazard assumption may be too restrictive.
- The **Weibull**, **Gompertz**, and **generalized gamma** models fit the Kaplan-Meier curve much better.

(d) To determine the hazard function for the fitted exponential model, you can run the command `summary(ex, type="hazard", ci=TRUE, tidy=FALSE)`. Can you figure out how to plot the estimated hazard functions for these four distributions on the same figure?

```
# hazard summary for the exponential model
# the full output is quite long so we only show 3 time points
summary(ex, type = "hazard", t=c(50,100,150), ci = TRUE, tidy = FALSE)
```

	time	est	lcl	ucl
1	50	0.002370928	0.002011819	0.002727452
2	100	0.002370928	0.002011819	0.002727452
3	150	0.002370928	0.002011819	0.002727452



- The exponential model has a **flat** hazard by construction.
- The other models allow the hazard to change over time, which is more realistic if the risk of death is not constant throughout follow-up.

(e) *Interpret the output.*

```
lungdat_ecog <- subset(lungdat, !is.na(ph.ecogcat))
we.reg <- flexsurvreg(
  Surv(time, status) ~ factor(ph.ecogcat),
  data = lungdat_ecog,
  dist = "weibull"
```

```
)
print_no_call(we.reg)

Estimates:
```

	data mean	est	L95%	U95%	se
shape	NA	1.3555	1.2005	1.5304	0.0839
scale	NA	555.7900	437.8351	705.5225	67.6455
factor(ph.ecogcat)1	0.4978	-0.2648	-0.5519	0.0224	0.1465
factor(ph.ecogcat)2	0.2247	-0.6824	-1.0074	-0.3573	0.1658

	exp(est)	L95%	U95%
shape	NA	NA	NA
scale	NA	NA	NA
factor(ph.ecogcat)1	0.7674	0.5758	1.0227
factor(ph.ecogcat)2	0.5054	0.3652	0.6995


```
N = 227, Events: 164, Censored: 63
Total time at risk: 69522
Log-likelihood = -1138.825, df = 4
AIC = 2285.651
```

term	beta	acceleration_factor	weibull_PH_hazard_ratio
factor(ph.ecogcat)1	-0.265	0.767	1.432
factor(ph.ecogcat)2	-0.682	0.505	2.522

- The Weibull AFT model uses ECOG 0 as the reference group.
- The estimated acceleration factor for ECOG 1 is 0.767, suggesting **shorter** survival compared with ECOG 0. However, its 95% confidence interval includes 1, so this effect is not clearly significant at the 5% level.
- The estimated acceleration factor for ECOG 2 is 0.505, with a 95% confidence interval entirely below 1, indicating significantly **shorter** survival compared with ECOG 0.
- The Weibull shape parameter is 1.356, which is greater than 1. This suggests that the baseline hazard **increases** over time.

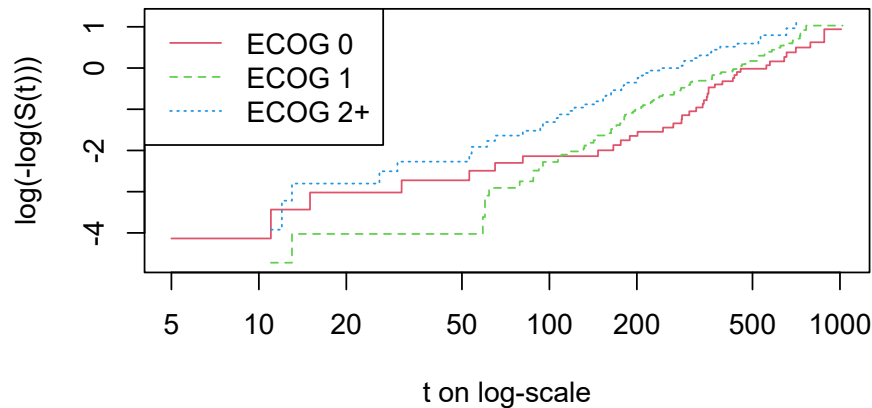
(f) *How would you interpret this output with regard to accelerating life and proportional hazards?*

- Since the Weibull model can also be interpreted in proportional hazards form, the corresponding hazard ratio is given by $\exp(-\hat{\alpha}\hat{\beta})$, where $\hat{\alpha}$ is the Weibull shape parameter and $\hat{\beta}$ is the AFT

coefficient.

- ECOG 1 has an estimated mortality hazard about 1.43 times that of ECOG 0, while ECOG 2 has an estimated mortality hazard about 2.52 times that of ECOG 0.
- **Worse** ECOG performance status is associated with shorter survival time and higher mortality risk.

(g) *What do you conclude from looking at this plot?*



- The complementary log-log curves show clear separation between the ECOG groups.
- The ECOG 2+ group tends to have a **higher** curve, suggesting a higher cumulative hazard and shorter survival compared with the ECOG 0 and ECOG 1 groups.
- The curves are *not* perfectly straight or perfectly parallel on the log-time scale, which suggests that the common-shape Weibull PH/AFT structure may only be an approximate working model.
- The non-parallel pattern indicates that ECOG may also affect the shape of the Weibull hazard function, not only the scale of survival time.
- Since the plot is based on KM step functions and the sample size becomes smaller at later times, some departures from parallelism are expected.
- The plot does not completely rule out the Weibull model, but it motivates checking a more flexible Weibull model in which the shape parameter is allowed to depend on ECOG.

(h) *What impact does ECOG have on the shape of the Weibull regression model just fitted?*

```

# ECOG 影响 scale 同时还影响 shape
we.reg.shape.ecog <- flexsurvreg(
  Surv(time, status) ~ factor(ph.ecogcat),
  anc = list(shape = ~ factor(ph.ecogcat)),
  data = lungdat_ecog,
  dist = "weibull"
)
ecog_shape_compare <- data.frame(
  model = c("Common shape", "Shape depends on ECOG"),
  logLik = c(we.reg$loglik, we.reg.shape.ecog$loglik),
  AIC = c(we.reg$AIC, we.reg.shape.ecog$AIC)
)
knitr::kable(ecog_shape_compare, digits = 3)

```

model	logLik	AIC
Common shape	-1138.825	2285.651
Shape depends on ECOG	-1137.741	2287.481

```
print_no_call(we.reg.shape.ecog)
```

Estimates:

	data mean	est	L95%	U95%	se
shape	NA	1.3178	1.0085	1.7220	0.1798
scale	NA	558.4727	435.3249	716.4574	70.9823
factor(ph.ecogcat)1	0.4978	-0.2669	-0.5558	0.0221	0.1474
factor(ph.ecogcat)2	0.2247	-0.7169	-1.0690	-0.3648	0.1797
shape(factor(ph.ecogcat)1)	0.4978	0.1161	-0.1999	0.4321	0.1612
shape(factor(ph.ecogcat)2)	0.2247	-0.0926	-0.4454	0.2603	0.1800
	exp(est)	L95%	U95%		
shape	NA	NA	NA		
scale	NA	NA	NA		
factor(ph.ecogcat)1	0.7658	0.5736	1.0224		
factor(ph.ecogcat)2	0.4883	0.3433	0.6944		
shape(factor(ph.ecogcat)1)	1.1231	0.8188	1.5405		
shape(factor(ph.ecogcat)2)	0.9116	0.6406	1.2973		

N = 227, Events: 164, Censored: 63

Total time at risk: 69522

```
Log-likelihood = -1137.741, df = 6
AIC = 2287.481
```

- The model allowing the Weibull **shape** parameter to depend on ECOG has a **slightly higher** log-likelihood than the common-shape model, but its AIC is **larger**. This means that the improvement in model fit is *not* large enough to justify the additional shape parameters.
- The estimated shape effects for ECOG 1 and ECOG 2+ are also not clearly significant, since their confidence intervals include 1 after exponentiation.
- There is no strong evidence that ECOG has an important effect on the **shape** of the Weibull hazard function.
- The simpler common-shape Weibull model is preferred. ECOG mainly appears to affect the **scale** of survival time, with worse ECOG performance associated with shorter survival, rather than changing the **shape** of the Weibull survival distribution.

(i) *What impact does weight loss in the last six months (wtlossyn variable) have on the risk and the shape of the Weibull regression model just fitted?*

```
lungdat_wt <- subset(lungdat, !is.na(ph.ecogcat) & !is.na(wtlossyn))
# ECOG 和 wtlossyn 只影响 scale
we.reg.wt <- flexsurvreg(
  Surv(time, status) ~ factor(ph.ecogcat) + factor(wtlossyn),
  data = lungdat_wt,
  dist = "weibull"
)
# ECOG 和 wtlossyn 影响 scale, 同时 wtlossyn 还影响 shape
we.reg.wt.shape <- flexsurvreg(
  Surv(time, status) ~ factor(ph.ecogcat) + factor(wtlossyn),
  anc = list(shape = ~ factor(wtlossyn)),
  data = lungdat_wt,
  dist = "weibull"
)
wt_compare <- data.frame(
  model = c("Weight loss affects scale", "Weight loss affects scale and shape"),
  logLik = c(we.reg.wt$loglik, we.reg.wt.shape$loglik),
  AIC = c(we.reg.wt$AIC, we.reg.wt.shape$AIC)
)
knitr::kable(wt_compare, digits = 3)
```

model	logLik	AIC
Weight loss affects scale	-1054.569	2119.138
Weight loss affects scale and shape	-1054.173	2120.346

```
print_no_call(we.reg.wt) # weight loss affects scale
```

Estimates:

	data mean	est	L95%	U95%	se
shape	NA	1.3825	1.2181	1.5690	0.0893
scale	NA	534.0497	401.4209	710.4990	77.7870
factor(ph.ecogcat)1	0.4977	-0.2665	-0.5589	0.0259	0.1492
factor(ph.ecogcat)2	0.2160	-0.6920	-1.0331	-0.3510	0.1740
factor(wtlossyn)1	0.7136	0.1091	-0.1552	0.3733	0.1348
	exp(est)	L95%	U95%		
shape	NA	NA	NA		
scale	NA	NA	NA		
factor(ph.ecogcat)1	0.7660	0.5718	1.0262		
factor(ph.ecogcat)2	0.5006	0.3559	0.7040		
factor(wtlossyn)1	1.1152	0.8563	1.4525		

N = 213, Events: 151, Censored: 62

Total time at risk: 67133

Log-likelihood = -1054.569, df = 5

AIC = 2119.138

```
print_no_call(we.reg.wt.shape) # weight loss affects scale and shape
```

Estimates:

	data mean	est	L95%	U95%	se
shape	NA	1.5149	1.2032	1.9072	0.1780
scale	NA	538.2675	410.3867	705.9974	74.4954
factor(ph.ecogcat)1	0.4977	-0.2739	-0.5630	0.0153	0.1475
factor(ph.ecogcat)2	0.2160	-0.7138	-1.0576	-0.3700	0.1754
factor(wtlossyn)1	0.7136	0.1105	-0.1407	0.3617	0.1282
shape(factor(wtlossyn)1)	0.7136	-0.1275	-0.4041	0.1490	0.1411
	exp(est)	L95%	U95%		
shape	NA	NA	NA		
scale	NA	NA	NA		
factor(ph.ecogcat)1	0.7604	0.5695	1.0154		

```
factor(ph.ecogcat)2      0.4898    0.3473    0.6908
factor(wtlossyn)1       1.1168    0.8687    1.4357
shape(factor(wtlossyn)1) 0.8803    0.6676    1.1607
```

```
N = 213, Events: 151, Censored: 62
Total time at risk: 67133
Log-likelihood = -1054.173, df = 6
AIC = 2120.346
```

- Here `wtlossyn = 1` means the patient lost weight in the last six months, while `wtlossyn = 0` means the patient did not lose weight or gained weight.
- After adjusting for ECOG performance score, `wtlossyn` does *not* appear to have a strong effect on the `scale` of the Weibull AFT model.
- In the common-shape model, the acceleration factor for `wtlossyn` is about 1.115, and its 95% confidence interval includes 1. This suggests that weight loss does *not* substantially shift the overall survival time `scale`.
- When `wtlossyn` is allowed to affect the Weibull `shape` parameter, the AIC **increases slightly** from 2119.138 to 2120.346. The estimated `shape` ratio for `wtlossyn` is about 0.880, with a 95% confidence interval from 0.668 to 1.161. This does not provide clear evidence that weight loss affects the `shape` of the hazard function.
- There is no strong evidence that weight loss in the last six months has an important effect on either the survival time `scale` or the Weibull hazard `shape`.

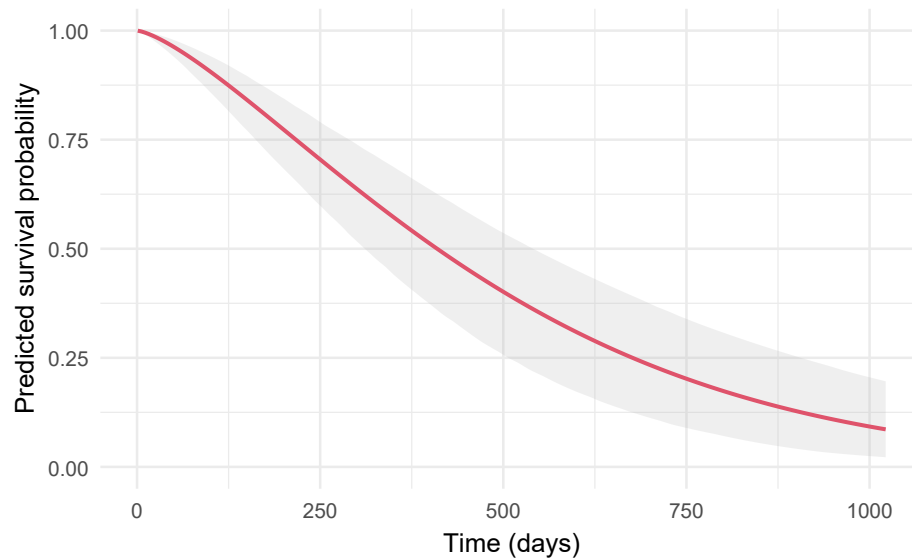
(j) *Can you figure out how to plot the predicted survival curve for an asymptomatic, advanced lung cancer patient who did not lose weight in the last six months?*

```
new.lung.patient <- data.frame(ph.ecogcat = 0, wtlossyn = 0) # 预测对象
pred.time <- seq(1, max(lungdat_wt$time, na.rm = TRUE), length.out = 200)
pred.surv <- summary(
  we.reg.wt, # common-shape model with ECOG and weight loss affecting scale
  type = "survival",
  t = pred.time,
  newdata = new.lung.patient,
  ci = TRUE,
  tidy = TRUE
)
```

est 表示生存概率

```
head(pred.surv)
```

	time	est	lcl	ucl	factor(ph.ecogcat)	factor(wtlossyn)
1	1.000000	0.9998305	0.9995387	0.9999434	0	0
2	6.130653	0.9979229	0.9955933	0.9991033	0	0
3	11.261307	0.9951921	0.9905083	0.9977501	0	0
4	16.391960	0.9919343	0.9848282	0.9960381	0	0
5	21.522613	0.9882690	0.9789233	0.9939661	0	0
6	26.653266	0.9842666	0.9724226	0.9916094	0	0



- The predicted survival curve is obtained for an asymptomatic advanced lung cancer patient with ECOG 0 and `wtlossyn` = 0.
- The model used for prediction is the common-shape Weibull model with ECOG and `wtlossyn` affecting the `scale` parameter.
- The fitted survival probability starts close to 1 at early times and decreases gradually over time.
- The predicted median survival time is roughly around 410 days, where the survival curve crosses 0.5.
- By around 1000 days, the predicted survival probability is about 0.09.
- The shaded region represents the 95% confidence interval, which becomes wider at later times, indicating greater uncertainty in long-term survival prediction.